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DATA PREPARATION & BASELINE

2,000 training questions (TriviaQA) + **200 held-out** for evaluation. Base model evaluated at Gen 0 to establish K_0 factual baseline per backbone.

Qwen 2.5 1.5B ($K_0=78$)

Gemma 3 1B ($K_0=46$)

Gemma 4 E2B ($K_0=76$)

2

RECURSIVE TRAINING (REPLACE-ONLY)

Each generation: fresh adapter trained on previous generation's synthetic outputs. **10 recursive generations**, strict data replacement (no accumulation).

QLoRA (4-bit NF4)

Full Fine-Tuning

3-5 seeds

3

PRESSURE MANIPULATION (5 AXES)

Systematic variation of effective training pressure along independent experimental axes.

Rank ($r=4$ to 256)

Learning Rate (10^{-6} to 2×10^{-5})

Method (QLoRA vs FFT)

Module Topology

Synthetic Exposure (0/10/25/50% removal)

4

EVALUATION

Per-generation measurement of factual retention, distributional shift, and adapter properties.

K_0 Retention

Effective Rank

Response Length

Distinct-n

MTLD

Stopword Ratio

5

REGIME CLASSIFICATION

Each configuration classified based on retention trajectory and distributional behavior.

Homeostatic (>90%)

Bounded (factual stable, distribution degrades)

Degradative (progressive loss)

6

PRESSURE-REGIME MAPPING

Effective Training Pressure identified as organizing principle: regime transitions are threshold-like within the tested grid, backbone-dependent, and controllable through multiple axes.